

Supply Chain – Marketing Analytics Dashboard

Using SQL + Power BI

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Executive Summary

BUSINESS BACKGROUND

Modern supply chain operations generate enormous amounts of transactional data every day. Without an integrated reporting system, managers often struggle to monitor operational performance, identify bottlenecks, and make timely business decisions. This project develops an interactive Supply Chain Analytics Dashboard using SQL and Power BI to transform raw transactional data into actionable business insights.

OBJECTIVES

1

Monitor business
performance

2

Track delivery efficiency

3

Evaluate product category
performance

4

Analyze customer purchasing
behavior

5

Support operational decision
making

EXPECTED VALUE

Instead of manually reviewing thousands of records, managers can quickly identify operational issues and business opportunities through interactive, self-service dashboards.

THE SCENARIO

A multinational retailer, five markets, one blind spot

Imagine a multinational sporting goods retailer operating across five global markets — LATAM, Europe, Pacific Asia, USCA, and Africa. The company has accumulated over 180,000 customer orders between 2015 and 2018.

Although operational data is available, managers face challenges answering fundamental performance questions using raw transactional records alone.

To address these challenges, an integrated analytical dashboard was developed.

MANAGERS CAN'T YET ANSWER

- 1 Which market contributes the most revenue?
- 2 Which departments generate the highest sales?
- 3 Are delivery operations meeting service level agreements?
- 4 Which customer segments should be prioritized?
- 5 Which product categories require inventory optimization?

WHAT THE TWO DASHBOARDS MUST ANSWER

Five questions, mapped to two dashboard pages

SUPPLY CHAIN DASHBOARD

Q1 Are delivery operations meeting service level agreements?

Q2 Which departments require the most operational attention?

MARKETING DASHBOARD

Q1 Which market contributes the most revenue?

Q2 Which departments generate the highest sales?

Q3 Which customer segments should be prioritized?

Which product categories require inventory optimization?

Answered jointly — inventory priority depends on both delivery reliability (Supply Chain) and revenue contribution (Marketing).

FROM RAW EXPORT TO DASHBOARD-READY TABLE

Dataset Overview

DATASET

DataCo Global Supply Chain Dataset

SCALE

180,519 Orders

TIME SPAN

2015 – 2018

RAW COLUMNS

30+ Variables

STRUCTURE

Single Fact Table

DOMAINS

Orders

Customer

Finance

Shipping

Products

Geography

DATA PREPARATION WORKFLOW



SQL TASKS PERFORMED

- 0 Removed invalid records
- 0 Standardized naming convention
- 0 Converted date fields to proper DATE type
- 0 Created a Quarter-Year time dimension
- 0 Optimized the table for analytical queries

Output: a single, query-ready fact table — the foundation for every KPI and visual in both dashboards.

KPI Definitions

Total Orders

DEMAND VOLUME

180,519

Total Sales

REVENUE

\$36.78M

Total Profit

PROFITABILITY

\$3.97M

Average Shipping Days

DELIVERY EFFICIENCY

3.50 days (actual)

Late Delivery Rate

SLA PERFORMANCE

54.9%

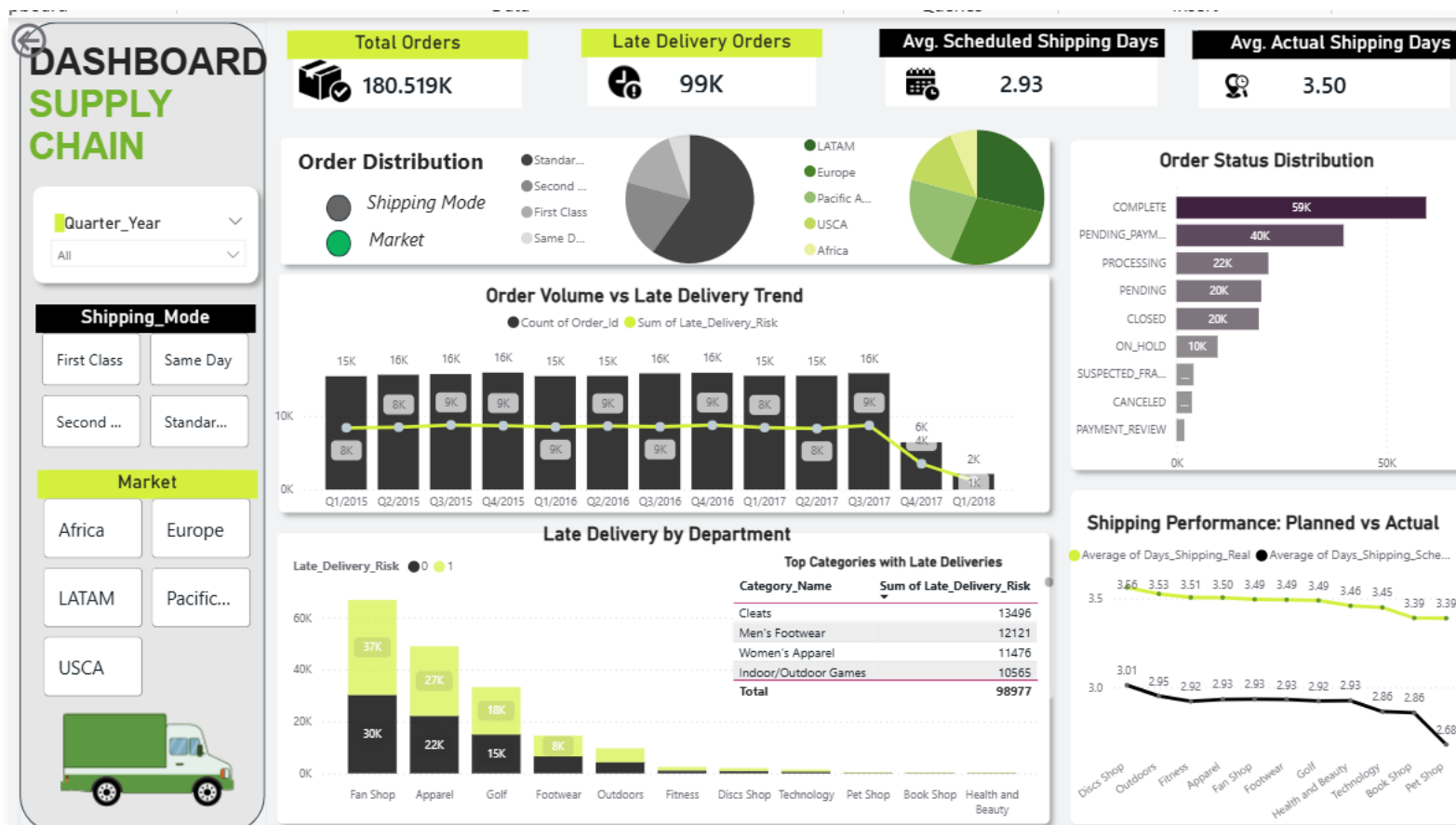
Average Order Value

CUSTOMER PURCHASING BEHAVIOR

\$203.75

Supply Chain Dashboard

High-level KPIs reveal the operational health of the network at a glance.

**180,519**

TOTAL ORDERS

99,000

LATE DELIVERY ORDERS

2.93 days

AVG. SCHEDULED SHIPPING

3.50 days

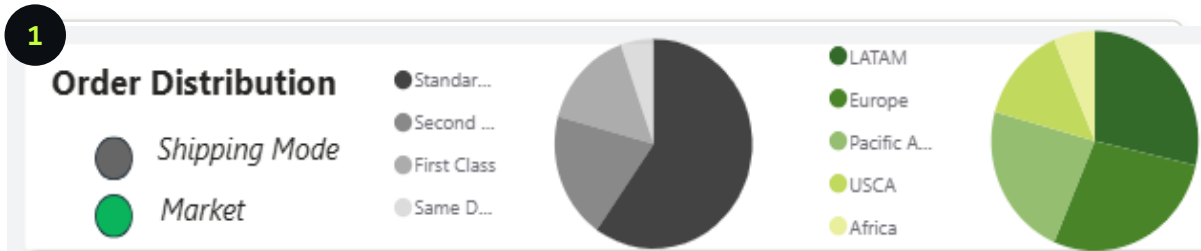
AVG. ACTUAL SHIPPING

Late delivery rate = $99,000 / 180,519 \approx 54.9\%$ — roughly one in every two orders ships later than promised.

WHERE VOLUME CONCENTRATES

Order distribution and lifecycle status

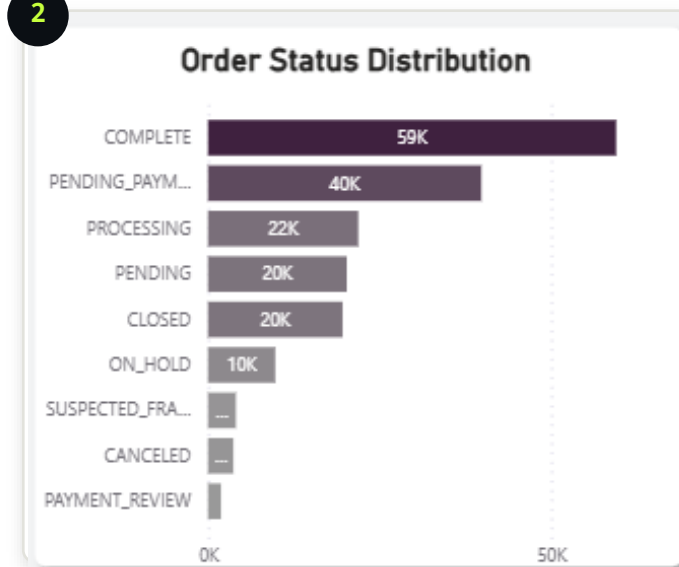
1



Order Distribution — Shipping Mode & Market

Orders are split across five shipping modes and five markets. Standard Class carries the largest share of volume, and LATAM and Europe together account for the majority of orders — the two markets to prioritize for capacity planning.

2



Order Status Distribution

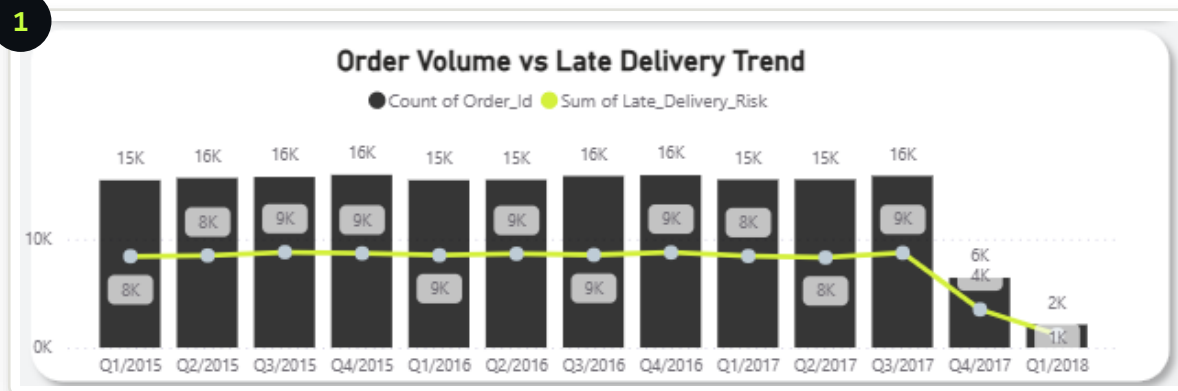
Only ~59K of 180,519 orders (~33%) sit in a fully Complete status. The remaining two-thirds are spread across Pending Payment (40K), Processing (22K), Pending (20K), Closed (20K) and On-Hold (10K) — a large working-in-progress pool worth monitoring.

READ THIS AS

Two structurally different problems: a shipping-mode/market concentration issue, and a status-pipeline backlog issue. Both need separate operational owners.

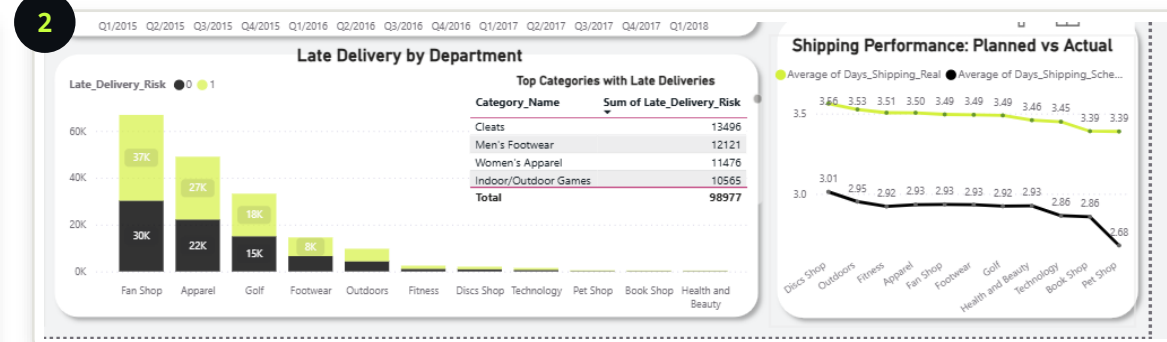
IS THE PROBLEM IMPROVING, AND WHERE DOES IT LIVE?

Order volume trend and department root-cause



Order Volume vs. Late Delivery Trend

Volume holds steady at ~15–16K orders/quarter from 2015 to Q3 2017, then drops sharply in the final two quarters (6K, then 2K) — most likely a data-completeness cut-off, not a genuine demand collapse.



Late Delivery by Department & Shipping Performance: Planned vs. Actual

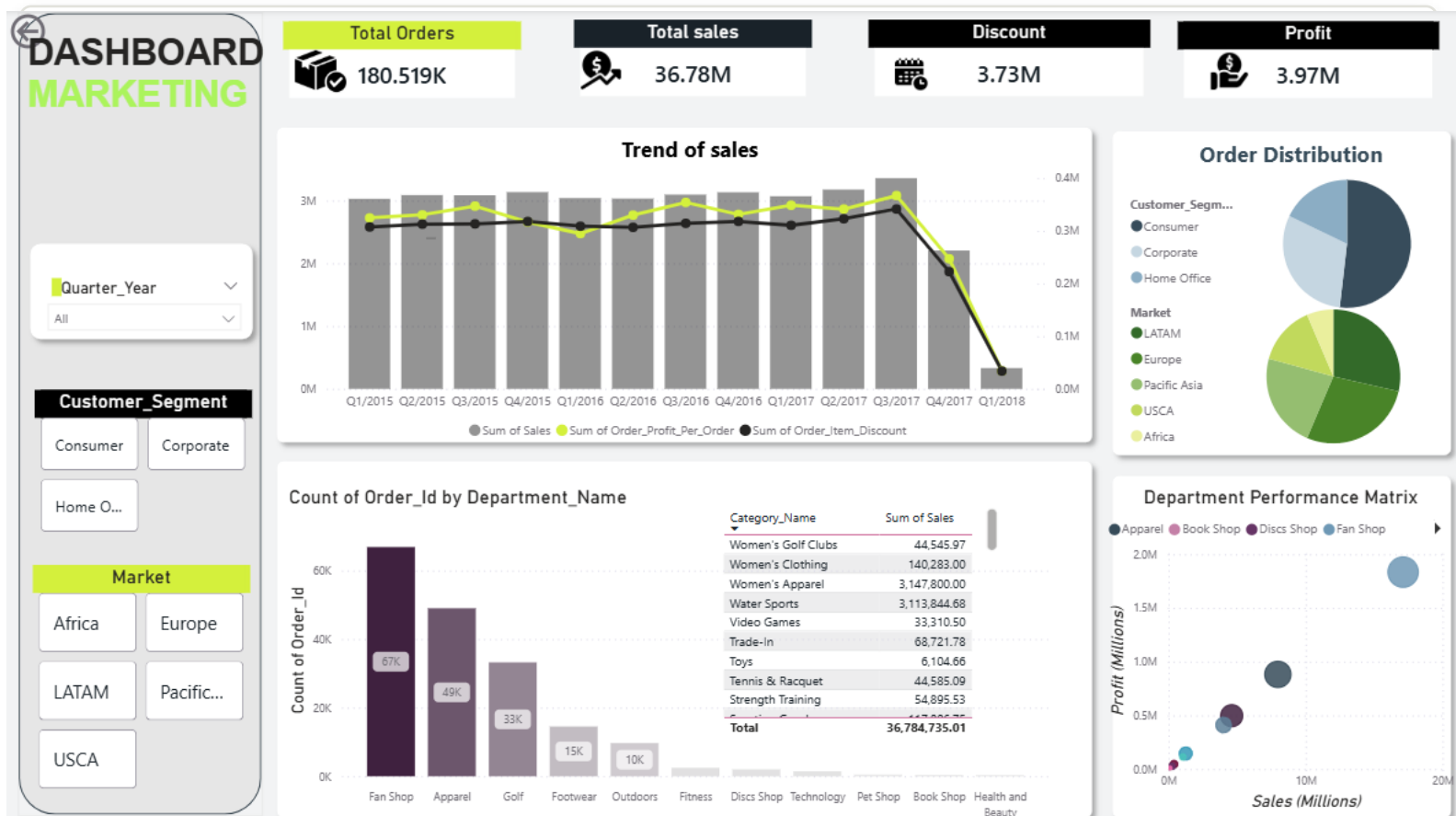
Late-delivery share is consistently ~55% across Fan Shop, Apparel and Golf alike — this is a network-wide scheduling issue, not a department-specific defect. The scheduled-vs-actual gap (~0.57 days) holds steady across all 13 quarters, with no improvement trend.

KEY TAKEAWAY

Late delivery is a systemic, network-wide scheduling problem — fixing it means re-baselining promised shipping days globally, not chasing individual categories.

Marketing Dashboard

The same order base, read through a revenue lens — sales, discount and profit.

**180,519**

TOTAL ORDERS

\$36.78M

TOTAL SALES

\$3.73M

DISCOUNT

\$3.97M

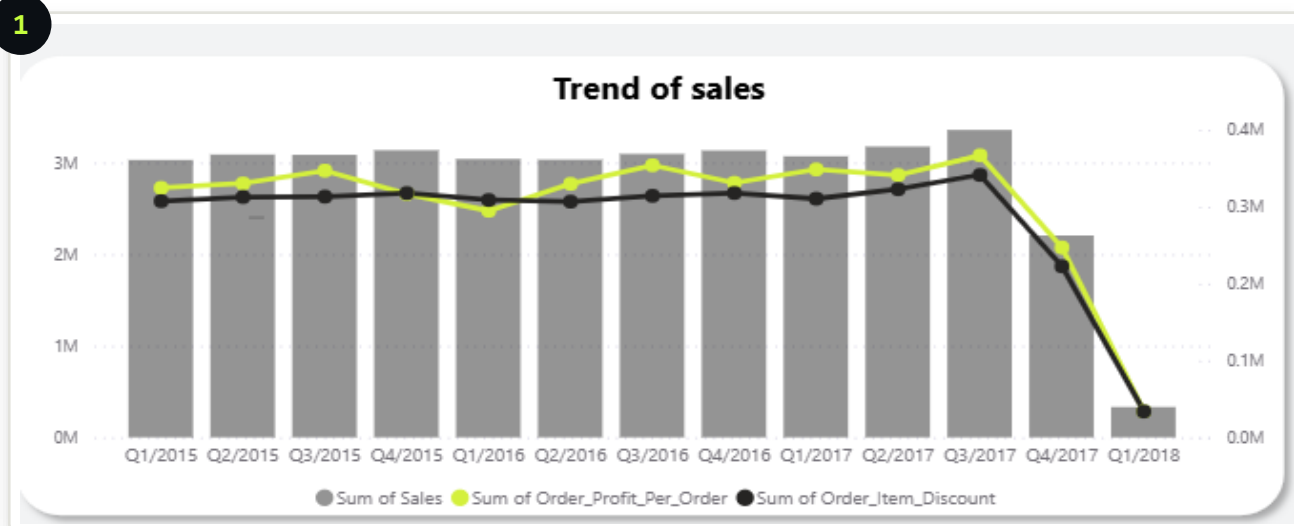
PROFIT

Profit margin $\approx 10.8\%$ · Discount rate $\approx 10.1\%$ · Avg. order value $\approx \$203.75$ — margin and discount move in a stable, 16 predictable band.

REVENUE OVER TIME, AND WHO IT COMES FROM

Sales trend, customer segments and markets

1



Trend of Sales — Sales, Profit & Discount by Quarter

Sales hold near \$2.5–3M/quarter through Q3 2017, with profit and discount tracking proportionally throughout — a stable margin, not a declining one. The final two quarters drop sharply, mirroring the same data-completeness cut-off seen on the Supply Chain page.

2



Order Distribution

Consumer is the largest customer segment; LATAM and Europe are the largest markets — consistent with the concentration seen in delivery volume.

REVENUE CONCENTRATION

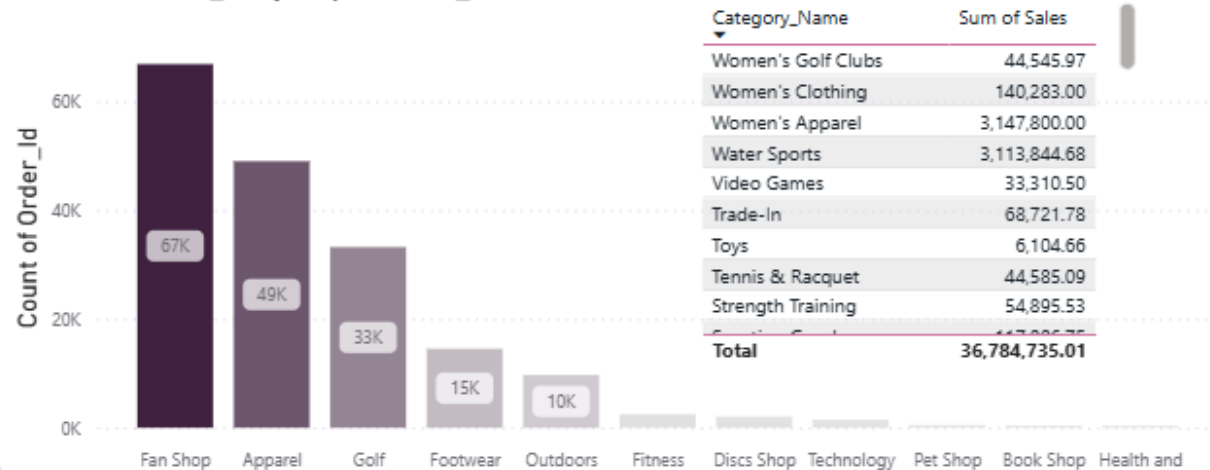
Women's Apparel and Water Sports alone contribute ≈\$6.26M of \$36.78M total sales (≈17%) — from just 2 of 15+ categories.

WHICH DEPARTMENTS ACTUALLY DRIVE THE BUSINESS

Order volume, category revenue and profit efficiency

1

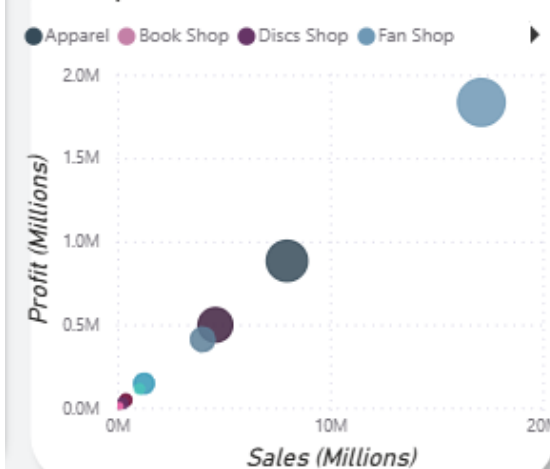
Count of Order_Id by Department_Name



Orders by Department & Category Revenue

2

Department Performance Matrix



Department Performance Matrix

ONE DEPARTMENT, OUTSIZED WEIGHT

Fan Shop drives 67K of 180,519 orders ($\approx 37\%$) and sits far above every other department on both sales ($\sim \$17\text{--}18\text{M}$) and profit ($\sim \$1.8\text{--}2.0\text{M}$) in the Department Performance Matrix. Apparel is a distant second ($\sim \$8\text{M}$ sales, $\sim \$0.9\text{--}1.0\text{M}$ profit); Book Shop and Discs Shop cluster near the origin — minimal contribution.

Business Insights

INSIGHT 1

Late delivery risk is systemic, not category-specific

OBSERVATION

Late-delivery share is uniform (~55%) across Fan Shop, Apparel and Golf alike.

EVIDENCE

Fan Shop: 37K/67K late · Apparel: 27K/49K late · Golf: 18K/33K late — the same ratio, regardless of department size.

BUSINESS IMPACT

The root cause sits at the scheduling-policy level, not within any single department or category.

Re-baseline Days_Shipping_Scheduled network-wide rather than tuning it department by department.

INSIGHT 2

The delivery gap has not closed in three years

OBSERVATION

Actual shipping consistently exceeds scheduled shipping across the full 2015–2018 window.

EVIDENCE

Scheduled ~2.93 days vs. Actual ~3.50 days — a ~0.57-day gap that holds steady across all 13 quarters.

BUSINESS IMPACT

Three years of data show no improvement trend, meaning the gap is structural, not seasonal.

Treat this as an SLA-design problem — revisit how shipping promises are calculated, not just how they're executed.

INSIGHT 3

The final two quarters are a data artifact, not a business decline

OBSERVATION

Order volume, sales and late-delivery counts all collapse in Q4/2017 and Q1/2018.

EVIDENCE

Orders drop from ~16K/quarter to 6K, then 2K; sales drop from ~\$3M to near \$0 — while the late-delivery ratio stays ~55% throughout.

BUSINESS IMPACT

A proportional drop across every metric points to partial data capture, not a genuine demand or quality collapse.

Validate data completeness for these two quarters before using them in any trend-based or board-level reporting.

Business Insights (continued)

INSIGHT 4

One department carries the business

OBSERVATION

Fan Shop dominates both order volume and financial performance.

EVIDENCE

Fan Shop: 67K/180,519 orders (≈37%), ~\$17–18M sales, ~\$1.8–2.0M profit — far above Apparel, the next largest (~\$8M sales).

BUSINESS IMPACT

A stockout or service failure in Fan Shop is a company-level risk, not a departmental one.

Give Fan Shop a dedicated stockout/service-continuity monitor; treat Book Shop, Discs Shop and similar small departments as low operational priority.

INSIGHT 5

Revenue concentrates in a handful of categories

OBSERVATION

A small number of categories generate a disproportionate share of total sales.

EVIDENCE

Women's Apparel (\$3.15M) and Water Sports (\$3.11M) alone contribute ≈17% of \$36.78M total sales; categories like Toys (\$6.1K) contribute under 0.02%.

BUSINESS IMPACT

Inventory and marketing effort spread evenly across categories is misallocated relative to where revenue actually comes from.

Apply ABC-style category tiering and prioritize inventory planning and campaign spend on the top revenue tier.

INSIGHT 6

Margin and discount are stable — a strength worth protecting

OBSERVATION

Profit and discount move proportionally with sales throughout the observed period.

EVIDENCE

Profit margin holds near 10.8% and discount rate near 10.1%, both consistent quarter over quarter with no erosion trend.

BUSINESS IMPACT

Unlike delivery performance, commercial economics are healthy today — but concentration risk (Insights 4–5) could destabilize this as the business scales.

Keep discount-to-margin monitoring active in the dashboard so future growth doesn't quietly erode today's stable economics.

Business Recommendations

Short-term

0 - 3 MONTHS

- Audit Q4/2017–Q1/2018 for data completeness before using them in trend reporting.
- Re-baseline scheduled shipping days network-wide given the uniform ~55% late rate.
- Stand up a Fan Shop stockout/service-continuity monitor.

Medium-term

3 - 12 MONTHS

- Run ABC revenue analysis and refocus inventory planning on top categories (Women's Apparel, Water Sports).
- Investigate carrier/route performance to close the persistent ~0.57-day scheduled-vs-actual gap.
- Build segment-aware marketing to grow Corporate and Home Office share, reducing Consumer concentration.

Long-term

1+ YEAR

- Implement demand forecasting so shipping capacity is planned ahead of demand, not reactive.
- Introduce predictive inventory planning tied to ABC category tiers.
- Integrate supplier/carrier performance monitoring directly into the Power BI model.

THANK YOU

From raw records to ready decisions.

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Supply Chain Analytics Dashboard · SQL + Power BI

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